



UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 5: Linear independence

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Slides: David Angell, Sean Gardiner, Laure Helme-Guizon and many others

An inconvenient truth (two really).

Example 1. Consider the set $S = \left\{ \vec{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$ and the vectors $\vec{u}_1 = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}, \vec{u}_2 = \begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix}$.

- (a) Can $\vec{u}_1$ be written as a linear combination of elements of S ? If so, is there more than one way to write $\vec{u}_1$ as a linear combination of elements of S ?
- (b) Can $\vec{u}_2$ be written as a linear combination of elements of S ? If so, is there more than one way to write $\vec{u}_2$ as a linear combination of elements of S ?



*Wouldn't it be nice to be in a situation where not only **any** vector could be written as a linear combination of some set of vectors but also the way of writing this linear combination would be **unique**?*

We could call the scalars in this linear combination “coordinates”.

Let our quest begin...

Linear independence

Definition (**Linear independence**)

Given a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ that is a subset of a vector space V with associated scalar field $\mathbb{F}$, the vectors in S are **linearly independent** if the **only** solution to

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0} \text{ where } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$$

is $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$. (This is sometimes known as **the trivial solution**.)

Note that $\lambda_1, \lambda_2, \dots, \lambda_n = 0$ is $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$ will always be a solution of the equation $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}$. The issue is whether there are *other* solutions.

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The vectors in S are **linearly dependent** if there is a non-trivial solution to the above equation – that is, if there is a solution to

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0} \text{ where } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$$

with at least one of the scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ not equal to zero.

Example 2. Are the vectors in the set $S = \left\{ \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$ linearly independent?

Worked solution.

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Worked solution. We wish to find all solutions to the equation

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system has a unique solution, which must be the trivial solution $\lambda_1 = \lambda_2 = \lambda_3 = 0$. So S is indeed a linearly independent set.

Notice we did not need the fourth column in the augmented matrix to reach this conclusion – the fact there is a unique solution is enough to confirm the vectors are linearly independent.

Can you remind me how to do it with Maple?

Example 2 (again). Are the vectors in the set $S = \left\{ \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$

linearly independent?

```
> with(LinearAlgebra):  
> A := <<1,4,3>|<2,3,1>|<1,-1,0>>; b:= <0,0,0>; Ab := <A|b>;
```

$$A := \begin{bmatrix} 1 & 2 & 1 \\ 4 & 3 & -1 \\ 3 & 1 & 0 \end{bmatrix}$$

$$b := \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ab := \begin{bmatrix} 1 & 2 & 1 & 0 \\ 4 & 3 & -1 & 0 \\ 3 & 1 & 0 & 0 \end{bmatrix}$$

```
> Uy := GaussianElimination(Ab);
```

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For example, one non-trivial solution to the equation is

$$\lambda_1 = 1, \lambda_2 = -1, \text{ and } \lambda_3 = 1.$$

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- Determining if vectors are linearly independent is really about solving a system of equations, so we can use augmented matrices and row-reduction.
- A small economy: the right hand side never changes (always zeros), so we need not write it down.
- We did not need to know *what* the solutions of the system are, only *how many* there are. We can get this immediately from the echelon form by noting that there is a non-leading column.

Maximum number of linearly independent vectors in $\mathbb{R}^3$?

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Answer.

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Answer. No, because doing the same sort of working as on the previous page will give a system of three equations in four unknowns. The echelon form might look like

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among other options (*exercise*: how many altogether?).

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Conclusion. We have shown that in $\mathbb{R}^3$, linear independent sets cannot have more than 3 elements.

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So, in certain vector spaces, there seem to be an upper bound on the number of elements an independent set can have.

Example 6. (a) Which vector(s) can be removed from the set

$$S = \left\{ \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \right\} \text{ to make } S \text{ a linearly independent set?}$$



It is given (Maple output) that the row echelon form of

$$(A|\mathbf{b}) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 0 \\ 4 & 3 & -1 & -2 & 0 \\ 3 & 1 & -2 & 4 & 0 \end{array} \right) \text{ is } (U|\mathbf{y}) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 0 \\ 0 & -5 & -5 & -6 & 0 \\ 0 & 0 & 0 & 7 & 0 \end{array} \right).$$

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Solution. The third column of U is non-leading, but if the corresponding vector had not been included in the augmented matrix then all columns of U would be leading and the row echelon form would imply the system has a unique solution. So if we exclude the **third** vector from S , we will get a linearly independent set of vectors.

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Similarly, notice that if instead the second column were excluded, the augmented matrix would again imply there is a unique solution. So the set S will also be linearly independent if we exclude the **second** vector instead.

So either $\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$ can be removed to ensure linear independence of S .

Example 6, continued. *Could we have done something else?*

Consider (again) the set $S = \left\{ \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \right\}$.

(b) Does removing the fourth vector make S a linearly independent set?



It is given (Maple output) that the row echelon form of

$$(A|\mathbf{b}) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 0 \\ 4 & 3 & -1 & -2 & 0 \\ 3 & 1 & -2 & 4 & 0 \end{array} \right) \text{ is } (U|\mathbf{y}) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 0 \\ 0 & -5 & -5 & -6 & 0 \\ 0 & 0 & 0 & 7 & 0 \end{array} \right).$$

Solution.

Linear independence problems as matrix problems

Theorem (**Linear independence**)

Given a field $\mathbb{F}$ and vector space $\mathbb{F}^m$ with a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, the vectors in S are linearly independent if and only if the row echelon form of the matrix $(\mathbf{v}_1 \mid \mathbf{v}_2 \mid \dots \mid \mathbf{v}_n)$ has a leading entry in every column.

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Notice this is equivalent to saying the system of equations $A\mathbf{x} = \mathbf{0}$, where $A = (\mathbf{v}_1 \mid \mathbf{v}_2 \mid \dots \mid \mathbf{v}_n)$, has a unique solution.

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Notice this is equivalent to saying that the system of equations $A\mathbf{x} = \mathbf{0}$, where $A = (\mathbf{v}_1 \mid \mathbf{v}_2 \mid \dots \mid \mathbf{v}_n)$, has infinitely many solutions.

Uniqueness of linear combinations

Theorem (**Uniqueness of linear combinations**)

Consider a vector space V with a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, and any vector $\mathbf{u} \in \text{span}(S)$, the vectors in set S are linearly independent if and only if the linear combination for $\mathbf{u}$ is unique – that is, there is only one way to write

$$\mathbf{u} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$$

for $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$.

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To see why this is true, we can prove both directions by contraposition. Since $\mathbf{u} \in \text{span}(S)$, we first write $\mathbf{u} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$ for some $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$.

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Supposing the vectors in set S are not linearly independent, then there exist $\mu_1, \mu_2, \dots, \mu_n \in \mathbb{F}$ not all zero such that $\mathbf{0} = \mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2 + \dots + \mu_n \mathbf{v}_n$. But adding $\mathbf{0}$ to the right-hand side of the equation for $\mathbf{u}$ yields

$$\mathbf{u} = (\lambda_1 + \mu_1) \mathbf{v}_1 + (\lambda_2 + \mu_2) \mathbf{v}_2 + \dots + (\lambda_n + \mu_n) \mathbf{v}_n,$$

which must be a different (non-unique) linear combination for $\mathbf{u}$.

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On the other hand, supposing $\mathbf{u}$ has more than one way to be written as a linear combination, we first write $\mathbf{u} = \mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2 + \dots + \mu_n \mathbf{v}_n$ for some $\mu_1, \mu_2, \dots, \mu_n \in \mathbb{F}$ where $\lambda_i \neq \mu_i$ for at least one value of i between 1 and n . But then

$$\mathbf{0} = \mathbf{u} - \mathbf{u} = (\lambda_1 - \mu_1) \mathbf{v}_1 + (\lambda_2 - \mu_2) \mathbf{v}_2 + \dots + (\lambda_n - \mu_n) \mathbf{v}_n$$

is some non-trivial linear combination for $\mathbf{0}$, implying S is linearly dependent.

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Given a vector space V with a set of vectors $S \subseteq V$, for any $\mathbf{v} \in V$, if S is linearly independent and $\mathbf{v} \notin \text{span}(S)$, then $S \cup \{\mathbf{v}\}$ is linearly independent.